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An analytical solution to the axisymmetric thermoelasticity problem for a cylinder with arbitrarily varying thermomechanical properties

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Abstract Analytical solutions to the axisymmetric heat-conduction and thermoelasticity problems for a long solid cylinder are constructed for arbitrary variation of the thermomechanical properties within the radial coordinate. By making use of the direct integration method, the steady-state heat-conduction equation is reduced to an integral equation of second kind. The numerical, analytical-numerical and analytical methods for solving the latter equation are suggested. The one-to-one relations between the stresses and displacements on the boundary are established, which allows for the reduction of the thermoelasticity problems with various types of the boundary conditions to the one in terms of stresses. The explicit expressions for the stress-tensor and displacement-vector components are obtained and analyzed for different types of inhomogeneity.

1 Introduction

Thermomechanical performance of structure members made of inhomogeneous materials is one of the most prominent problems of solid mechanics in the recent decades. Among a number of bodies of canonical shape, the bodies of revolution and, particularly, solid and hollow cylinders, are a main focus of interest due to their wide engineering applications and, on the other hand, the possibility for reducing the general three-dimensional problems to two- or one-dimensional ones by taking advantage of their shape symmetry. The major part of studies on radially inhomogeneous cylinders rest upon the hypotheses of plane stresses or plane strains, where the stresses and displacements are assumed to be independent of the axial coordinate (see, e.g., [34,37] for extensive reviews of studies).

In contrast to the plane problems, the two-dimensional axisymmetric problems, when both radial and axial variations of the involved functions are taken into consideration, for inhomogeneous materials have received less attention due to the complexity of the governing equations. The majority of available solutions have been constructed for particular cases of inhomogeneity. For instance, the axisymmetric elasticity and thermoelasticity problems were treated in [18] for the Young's modulus given by a power function of the radial coordinate. In this case, the exact solutions of the Navier equations were constructed in the Fourier mapping space. The same type of inhomogeneity was addressed in [19] for the analysis of axisymmetric thermal stresses in a short hollow cylinder with simply supported or fixed end-faces. In [20] and [24], the

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same approach was adopted for the analysis of three-dimensional problems for finite-length cylinders. A numerical solution for the case of finite- and infinite-length cylinders was constructed in [13] by means of a mesh-free method. Results were given for the cases of power and generalized power dependence of the Young's modulus on the radial coordinate. The dynamic thermal stresses in a finite-length hollow cylinder due to the thermal shock were analyzed in [27]. The power-law variation of the material properties within both radial and axial coordinate was considered in [6] and [7] with concern to the natural vibration modes of a finite-length hollow cylinder. A numerical-analytical solution to a multi-field problem for a double-layer finite-length hollow cylinder with power-law variation of the material properties was given in [11]. A solution to the three-dimensional problem for a fiber-reinforced cylindrical panel with simply supported edges was constructed in [40] under the assumption of a linear variation of the reinforcement volume fraction in the radial direction by making use of a semi-analytical procedure based on the differential quadrature method. In [3,4] and [5], this approach was extended to the case of an orthotropic panel with power-law variation of the volume fraction within the radial coordinate. The general cases of volume fraction distributions were addressed in [2]. Detailed reviews on the recent development in the analysis of inhomogeneous materials with material properties in the form of power, exponential, or sigmoid functions are given, e.g., in [9,12,15,21].

Having treated a radially inhomogeneous hollow cylinder by making use of the concept of a multilayer solid, Kim and Noda [23] employed the method of Green function in order to solve the thermoelasticity problem in the case of unsteady axisymmetric thermal loading. The same concept has been used in [28] for the analysis of axisymmetric thermal stresses in a cylinder of finite length. The axisymmetric thermo-electro-mechanical problem of an FGM cylinder, which is rigidly confined between the inner and outer piezoelectric cylindrical layers, was treated in [1] for the case of simple power rule for the variation of conductivity with the radius.

The advancement in modeling and analysis of structural elements with spatial variation of the material properties calls for the development of efficient methods applicable for solving the relevant heat-conduction, elasticity, and thermoelasticity problems with arbitrary profile of inhomogeneity under different-type complex loadings. Approximate analytical solutions of the heat-conduction and thermoelasticity problems for a hollow cylinder with arbitrary dependence of the material properties on the radial coordinate were given in [38] by means of an iterative power-series solution procedure. An analytical solution of the thermoelasticity problem for a solid cylinder with arbitrary radial variations of the material properties was recently presented in [33] by making use of a direct method based on the reduction to integral equations with the subsequent employment of the resolvent-kernel solution technique. However, there is a strong need for analytical solutions explicitly relating the inlet and outlet data for arbitrary dependences of the material properties on the spatial coordinates. This need is dictated, e.g., by the problems of optimal design of inhomogeneous (or FGM) structures [8], optimal control of the thermal or stress-strain regimes [41]. On the other hand, the analysis of thermomechanical performance of arbitrarily inhomogeneous materials presents a challenge for both analytical and numerical modes of attack due to significant theoretical and practical complications and calls for a number of simplifying assumptions, which are usually accepted in the relevant literature but are potentially insufficient [25].

The intent of this paper aims to obtain analytical solutions to the axisymmetric heat-conduction and thermoelasticity problems for a long solid cylinder exposed to external thermal and force loadings and internal heat sources. We restrict our attention to the thermal effect, which is assumed to be the dominant factor in the considered problems, and, thus, we assume only the thermophysical properties (the linear thermal expansion coefficient and the heat-conduction coefficient) of the cylinder to be arbitrary functions of the radial coordinate. When solving the heat-conduction problem, the thermal loadings are assumed to be variable along the axial coordinate. By making use of the direct integration method [31], the formulated heat-conduction problem has been reduced to an integral equation of the second kind, which can be solved by making use of various analytical or numerical techniques. By means of the resolvent-kernel technique [32], for instance, the solution was given in a closed analytical form, which was efficiently used for the determination of thermal stresses and displacements arising in the cylinder due to the determined temperature distribution. Using the same technique [31], the original thermoelasticity equations were reduced to two governing equations for two key functions. Then the boundary conditions were transformed into the set of boundary and integral conditions for the key functions. Having solved the obtained problems, we managed to determine the stress-tensor components by their expressions through the key functions, which have been obtained from the equilibrium equations. As a result, the solution appeared in the form of functional dependences on the given thermal loadings and boundary tractions. Due to the latter reason, the elastic displacements have been computed directly by means of Cauchy integration along with the use of constitutive equations. The one-to-one relations between the stress-tensor and displacement-vector components on the boundary have been established, which allow for the reduction of the boundary conditions in terms of displacements, or the mixed ones (when both stresses and displacements are given on the boundary) to the boundary conditions in terms of stresses, for those the problem has already been solved. On the basis of the obtained solution, the numerical analysis is carried out.

2 Formulation of the problem

Consider a long solid cylinder of radius R related to the cylindrical-polar coordinate system (r, φ, ζ) so that the origin O lies on its main axis, which coincides with the coordinate axis $O\zeta$. Assume the cylinder to be made of an inhomogeneous material exhibiting variation of its thermophysical properties within the radial coordinate. The cylinder is subjected to an axisymmetric temperature field $T(\rho, z)$, which is found from the following heat-conduction equation [17]:

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\lambda(\rho)\frac{\partial T(\rho, z)}{\partial \rho}\right) + \lambda(\rho)\frac{\partial^2 T(\rho, z)}{\partial z^2} = \Xi(\rho, z), \tag{1}$$

which is accompanied by the following generalized boundary condition:

$$\left(aT(\rho, z) + b\lambda(\rho)\frac{\partial T(\rho, z)}{\partial \rho}\right)\bigg|_{\rho=1} = T_0(z) \tag{2}$$

and the finiteness condition

$$|T(0, z)| < \infty. \tag{3}$$

Here, $\rho = r/R$, $z = \zeta/R$, $\lambda(\rho)$ is the heat-conduction coefficient, which is an arbitrary function of the coordinate ρ and $\lambda(\rho) \neq 0$, $\forall \rho \in [0, 1)$, $\Xi(\rho, z)$ is the density of internal heat sources, $T_0(z)$ is a given function. Note that if $b = 0$, then the boundary condition (2) is of the Dirichlet type (the first-kind boundary condition) and presents the distribution of temperature on the lateral surface of the cylinder. If we set $a = 0$ in (2), then we have to deal with the Neumann-type boundary condition (the second-kind boundary condition) presenting the radial heat flux through the lateral surface. The case of $a \neq 0$ and $b \neq 0$ covers the boundary condition of heat exchange through the lateral surface.

The elastic stresses, which are excited by the temperature field $T(\rho, z)$ and the force loadings applied to the lateral surface, can be determined from the equations of equilibrium [16]

$$\frac{\partial}{\partial \rho}(\rho\sigma_r) + \rho\frac{\partial \sigma_{rz}}{\partial z} = \sigma_\varphi, \quad \frac{\partial}{\partial \rho}(\rho\sigma_{rz}) + \rho\frac{\partial \sigma_z}{\partial z} = 0 \tag{4}$$

along with the boundary conditions

$$\sigma_r(1, z) = -p(z), \qquad \sigma_{rz}(1, z) = q(z). \tag{5}$$

Here, σ_r, σ_φ, σ_z and σ_{rz} are the normal and shearing stress-tensor components, and $p(z)$ and $q(z)$ are given functions. The normal, ε_r, ε_φ, ε_z, and shearing, ε_{rz}, strains are related to the stress field by the following constitutive equations

$$\begin{aligned} E\varepsilon_r =& \sigma_r - \nu\left(\sigma_\varphi + \sigma_z\right) + \alpha(\rho)ET, \quad E\varepsilon_\varphi = \sigma_\varphi - \nu\left(\sigma_r + \sigma_z\right) + \alpha(\rho)ET, \\ E\varepsilon_z =& \sigma_z - \nu\left(\sigma_r + \sigma_\varphi\right) + \alpha(\rho)ET, \quad E\varepsilon_{rz} = 2(1+\nu)\sigma_{rz}, \end{aligned} \tag{6}$$

where E and ν are the constant Young's modulus and Poisson ratio, respectively, $\alpha(\rho)$ is the linear thermal expansion coefficient, which is an arbitrary function of the radial coordinate. The strain-tensor components are expressed through the radial, u_r, and axial, u_z, displacements as

$$\varepsilon_r = \frac{\partial u_r}{\partial \rho}, \quad \varepsilon_\varphi = \frac{u_r}{\rho}, \quad \varepsilon_z = \frac{\partial u_z}{\partial z}, \quad \varepsilon_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial \rho}. \tag{7}$$

In the case when the force loadings on the lateral surface remain unknown, but, instead, the boundary displacements are given, then condition (5) should be substituted with the following one:

$$u_r(1, z) = u(z), \qquad u_z(1, z) = w(z), \tag{8}$$

where $u(z)$ and $w(z)$ are known functions. In both cases, the finiteness conditions

$$|\sigma_r(0,z)| < \infty, \qquad |\sigma_{rz}(0,z)| < \infty, \qquad |u_r(0,z)| < \infty, \qquad |u_z(0,z)| < \infty \tag{9}$$

have to be fulfilled.

Our intent is to develop a technique for the solution of Eq. (1) with conditions (2) and (3) in order to determine the static temperature $T(\rho, z)$ in the cylinder. Then, we will solve the thermoelasticity problem (4)–(7) in order to find the thermal stresses and displacements due to the temperature $T(\rho, z)$ and boundary tractions (5) or boundary displacements (8) under the finiteness conditions (9).

3 Solution construction

3.1 Determination of the stationary temperature field

In order to separate the variables in Eq. (1) and solve the problem (1)–(3), we employ the Fourier integral transform

$$\overline{f(\rho; s)} \equiv \bar{f}(\rho) = \int_{-\infty}^{\infty} f(\rho, z)\exp(-isz)dz, \tag{10}$$

where $i^2 = -1$, s is the parameter of integral transform and f is a function, for which the transform (10) exists in a common or generalized meaning [10]. By making use of the transform (10), the problem (1)–(3) can be given in the form

$$\frac{d^2\bar{T}(\rho)}{d\rho^2} + \left(\frac{1}{\rho} + \frac{1}{\lambda(\rho)}\frac{d\lambda(\rho)}{d\rho}\right)\frac{d\bar{T}(\rho)}{d\rho} - s^2\bar{T}(\rho) = \frac{\bar{\Xi}(\rho)}{\lambda(\rho)}, \tag{11}$$

$$|\bar{T}(0)| < \infty, \tag{12}$$

$$\left.\left(a\bar{T}(\rho) + b\lambda(\rho)\frac{d\bar{T}(\rho)}{d\rho}\right)\right|_{\rho=1} = \bar{T}_0. \tag{13}$$

By taking the formula $\dfrac{1}{\lambda(\rho)}\dfrac{d\lambda(\rho)}{d\rho} = \dfrac{d\ln(\lambda(\rho))}{d\rho}$ into account, Eq. (11) can be given in the form

$$\frac{d^2\bar{T}(\rho)}{d\rho^2} + \frac{1}{\rho}\frac{d\bar{T}(\rho)}{d\rho} - s^2\bar{T}(\rho) = \frac{\bar{\Xi}(\rho)}{\lambda(\rho)} - \frac{d\ln(\lambda(\rho))}{d\rho}\frac{d\bar{T}(\rho)}{d\rho}.$$

With condition (12) in mind, the solution to the latter equation can be given as [26]

$$\bar{T}(\rho) = \bar{C}I_0(s\rho) + \frac{1}{2}\int_0^1 \eta\left(\frac{\bar{\Xi}(\eta)}{\lambda(\eta)} - \frac{d\ln(\lambda(\eta))}{d\eta}\frac{d\bar{T}(\eta)}{d\eta}\right)\chi_{00}(\rho,\eta)\mathrm{sgn}(\rho-\eta)d\eta,$$

where

$$\chi_{lm}(\rho,\eta) = I_l(s\rho)K_m(s\eta) - (-1)^{l+m}K_l(s\rho)I_m(s\eta), \tag{14}$$

$I_l(s\rho)$ and $K_m(s\rho)$ are the first- and second-kind modified Bessel functions of order l and m, respectively, l and m are integers, $\bar{C}$ is a constant of integration, and

$$\mathrm{sgn}(\rho-\eta) = \begin{cases} 1, & \rho > \eta, \\ -1, & \rho < \eta. \end{cases}$$

After integration of the second term in the integrand of the latter equation, it can be given in the following form:

$$\bar{T}(\rho) = CI_0(s\rho) + Q(\rho) + \int_0^1 \bar{T}(\eta)\mathcal{K}(\rho,\eta)d\eta, \tag{15}$$

where

$$C=\bar{C}+\frac{\bar{T}(1)}{2}\frac{d\ln(\lambda(1))}{d\rho}K_0(s),\quad Q(\rho)=\frac{1}{2}\int\limits_0^1\eta\frac{\bar{\Xi}(\eta)}{\lambda(\eta)}\chi_{00}(\rho,\eta)\mathrm{sgn}(\rho-\eta)d\eta,$$
$$\mathcal{K}(\rho,\eta)=\frac{1}{2}\frac{d}{d\eta}\left(\eta\frac{d\ln(\lambda(\eta))}{d\eta}\chi_{00}(\rho,\eta)\right)\mathrm{sgn}(\rho-\eta).$$

Substituting (15) into boundary condition (13) yields the integral equation

$$\bar{T}(\rho)=\gamma\bar{T}_0I_0(s\rho)+\Theta(\rho)+\int\limits_0^1\bar{T}(\eta)\mathcal{L}(\rho,\eta)d\eta,\tag{16}$$

where

$$\Theta(\rho)=\frac{1}{2}\int\limits_0^\rho\eta\frac{\bar{\Xi}(\eta)}{\lambda(\eta)}\left(\chi_{00}(\rho,\eta)\mathrm{sgn}(\rho-\eta)-I_0(s\rho)\varphi(\eta)\right)d\eta,$$
$$\mathcal{L}(\rho,\eta)=\frac{1}{2}\frac{d}{d\eta}\left(\eta\frac{d\ln(\lambda(\eta))}{d\eta}\chi_{00}(\rho-\eta)\right)\mathrm{sgn}(\rho-\eta)-\frac{I_0(s\rho)}{2}\frac{d}{d\eta}\left(\eta\frac{d\ln(\lambda(\eta))}{d\eta}\varphi(\eta)\right),$$
$$\varphi(\eta)=\left(a\chi_{00}(1,\eta)+bs\lambda(1)\chi_{10}(1,\eta)\right)\gamma,\quad\gamma=\left(aI_0(s)+bs\lambda(1)I_1(s)\right)^{-1}.$$

Equation (16) is an integral equation of the second kind which always has a unique solution [14]. This solution can be achieved by either analytical or numerical means [32]. In [29], an analogous equation has been solved by means of the successive approximation technique. A numerical approach resting upon the use of quadrature formulae has been presented, e.g., in [22]. In [33], the method of resolvent-kernel has been employed in order to achieve an explicit form analytical solution. Below, we present the solutions of Eq. (16) by means of the available methods.

3.1.1 A numerical solution constructed by means of the quadrature formulae

In order to employ the quadrature formulae, we consider a uniform element discretization of the interval [0, 1] as $\rho_0=0,\rho_1,\rho_2,...,\rho_{M-1},\rho_M=1$, where $\rho_j=\rho_0+hj$, $h=1/M$, $j=1,2,...,M$, $M>0$ is an integer. With the introduced partition in view, we can construct a discrete analogue for the integral in Eq. (16) by making use of an appropriate quadrature formula. Consider, for example, the trapezoid rule

$$\int\limits_0^1\bar{T}(\eta)\mathcal{L}(\rho,\eta)d\eta=h\left(\frac{1}{2}\bar{T}(\rho_M)\mathcal{L}(\rho,\rho_M)+\frac{1}{2}\bar{T}(\rho_0)\mathcal{L}(\rho,\rho_0)+\sum_{j=1}^{M-1}\bar{T}(\rho_j)\mathcal{L}(\rho,\rho_j)\right).$$

With this formula in view, Eq. (16) can be reduced to the following set of linear algebraic equations:

$$\begin{pmatrix}c_{00},c_{01},\ldots,c_{0M}\\c_{10},c_{11},\ldots,c_{1M}\\\vdots\quad\ddots\quad\vdots\\c_{M0},c_{M1},\ldots,c_{MM}\end{pmatrix}\begin{pmatrix}\bar{T}(\rho_0)\\\bar{T}(\rho_1)\\\vdots\\\bar{T}(\rho_M)\end{pmatrix}=\begin{pmatrix}d_0\\d_1\\\vdots\\d_M\end{pmatrix}.\tag{17}$$

Here,

$$c_{00}=\mathcal{L}(\rho_0,\rho_0)-\frac{2}{h},\ c_{0j}=2\mathcal{L}(\rho_0,\rho_j),\ c_{0M}=\mathcal{L}(\rho_0,\rho_M),\ d_0=-\frac{2}{h}\left(\gamma\bar{T}_0+\Theta(\rho_0)\right),\ c_{m0}=\mathcal{L}(\rho_m,\rho_0),$$
$$c_{mj}=\frac{2}{h}\left(\delta_{jm}-h\mathcal{L}(\rho_m,\rho_j)\right),\ c_{mM}=\mathcal{L}(\rho_m,\rho_M),\ d_m=-\frac{2}{h}\left(\gamma\bar{T}_0I_0(s\rho_m)+\Theta(\rho_m)\right),\ c_{M0}=\mathcal{L}(\rho_M,\rho_0),$$
$$c_{Mj}=2\mathcal{L}(\rho_M,\rho_j),\ c_{MM}=\mathcal{L}(\rho_M,\rho_M)-\frac{2}{h},\ d_M=-\frac{2}{h}\left(\gamma\bar{T}_0I_0(s)+\Theta(\rho_M)\right),\ \{j,m\}=1,2,...,M-1,$$

and δ_{jm} is the Kronecker delta. Having determined the set of values $\bar{T}(\rho_j)$, $j = 0, 1, 2, \ldots, M$, from system (17) for discrete values of the parameter s, we can then use the discrete analogue of the Fourier inverse transform [10]

$$f(\rho, z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{f}(\rho) \exp(isz) ds \tag{18}$$

in order to compute the temperature in the physical domain for the values $\rho_0, \rho_1, \ldots, \rho_M$.

3.1.2 An analytical-numerical solution by means of the successive approximations

An analytical-numerical solution to Eq. (16) can be constructed in the form [29]

$$\bar{T}(\rho) = \lim_{k\to\infty} \bar{T}^{\{k\}}(\rho), \tag{19}$$

where

$$\bar{T}^{\{0\}}(\rho) = \gamma \bar{T}_0 I_0(s\rho) + \Theta(\rho), \quad \bar{T}^{\{k\}}(\rho) = \gamma \bar{T}_0 I_0(s\rho) + \Theta(\rho) + \int_0^1 \bar{T}^{\{k-1\}}(\eta)\mathcal{L}(\rho, \eta)d\eta, \quad k > 0.$$

Due to the fact that the initial guess $\bar{T}^{\{0\}}$ is found as an exact solution of Eq. (16) without the integral constituent effects in the very rapid convergence of this method (see, e.g., [37]). After the limit (19) is found, the temperature can be restored in the physical domain on the basis of formula (18) by either analytical or numerical means.

3.1.3 An analytical resolvent-kernel solution

Having applied the resolvent-kernel technique [35], the solution to Eq. (16) can be given in the form

$$\bar{T}(\rho) = \gamma \bar{T}_0 \psi(\rho) + \theta(\rho), \tag{20}$$

where

$$\psi(\rho) = I_0(s\rho) + \int_0^1 I_0(s\eta)\mathcal{R}(\rho, \eta)d\eta, \quad \theta(\rho) = \Theta(\rho) + \int_0^1 \Theta(\eta)\mathcal{R}(\rho, \eta)d\eta, \tag{21}$$

$$\mathcal{R}(\rho, \eta) = \sum_{n=0}^{\infty} \mathcal{R}_{n+1}(\rho, \eta), \quad \mathcal{R}_1(\rho, \eta) = \mathcal{L}(\rho, \eta), \quad \mathcal{R}_{n+1}(\rho, \eta) = \int_0^1 \mathcal{R}_1(\rho, \tau)\mathcal{R}_n(\tau, \eta)d\tau. \tag{22}$$

By taking into consideration the diminution of the recurring kernels $\mathcal{R}_n(\rho, \eta)$ by both arguments with growing number n, the series (22) can be truncated for practical computations. Then the solution (20) can be computed with formulae (21), where the resolvent (22) is substituted with

$$\mathcal{R}(\rho, \eta) \approx \mathcal{R}^{\{N\}}(\rho, \eta) = \sum_{n=0}^{N} \mathcal{R}_{n+1}(\rho, \eta). \tag{23}$$

Here, N is a positive integer, which allows for the satisfaction of Eq. (16) within the required accuracy. Parameter N can be determined by either the sequence of numerical experiments or the analytical procedure [30]. In the physical domain, the temperature can be computed by putting (20) into (18).

3.2 Solution of the thermoelasticity problem

3.2.1 Determination of thermal stresses and displacements due to the found temperature and external force loadings

By eliminating the displacements from the Cauchy equation (7), we can obtain the following classical compatibility conditions [16]

$$\rho\frac{\partial\varepsilon_\varphi}{\partial\rho}=\varepsilon_r-\varepsilon_\varphi,\qquad \rho\frac{\partial^2\varepsilon_\varphi}{\partial z^2}-\frac{\partial\varepsilon_{rz}}{\partial z}+\frac{\partial\varepsilon_z}{\partial\rho} \tag{24}$$

in terms of strains. Making use of the constitutive Eq. (6) allows for the representation of (24) in terms of stresses as follows:

$$\begin{aligned}&\rho\frac{\partial}{\partial\rho}\left(\sigma_\varphi-\nu\left(\sigma_r+\sigma_z\right)+\alpha(\rho)ET\right)+(1+\nu)\left(\sigma_\varphi-\sigma_r\right)=0,\\&\rho\frac{\partial^2}{\partial z^2}\left(\sigma_\varphi-\nu\left(\sigma_r+\sigma_z\right)+\alpha(\rho)ET\right)-2\left(1+\nu\right)\frac{\partial\sigma_{rz}}{\partial z}+\frac{\partial}{\partial\rho}\left(\sigma_z-\nu\left(\sigma_r+\sigma_\varphi\right)+\alpha(\rho)ET\right)=0.\end{aligned} \tag{25}$$

By following the solution strategy [33,36,39], Eqs. (4) and (25) can be reduced to the two governing equations

$$\Delta_1\sigma_{rz}(\rho,z)=-\frac{1}{1+\nu}\frac{\partial^2}{\partial\rho\partial z}\left(\sigma(\rho,z)+\alpha(\rho)ET(\rho,z)\right),\quad \Delta_2\left(\frac{1-\nu}{E}\sigma(\rho,z)+2\alpha(\rho)T(\rho,z)\right)=0 \tag{26}$$

along with the expressions

$$\rho^2\sigma_r(\rho,z)=\int\limits_0^\rho\eta\left(\sigma(\eta,z)-\sigma_z(\eta,z)-\eta\frac{\partial\sigma_{rz}(\eta,z)}{\partial z}\right)d\eta, \tag{27}$$

$$2\sigma_z(\rho,z)=-\int\limits_{-\infty}^{\infty}\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\sigma_{rz}(\rho,\zeta)\right)\operatorname{sgn}(z-\zeta)d\zeta. \tag{28}$$

Here, $\Delta_1=\frac{\partial}{\partial\rho}\left(\frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho)\right)+\frac{\partial^2}{\partial z^2},\quad \Delta_2=\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}\right)+\frac{\partial^2}{\partial z^2}$ and

$$\sigma=\sigma_r+\sigma_\varphi+\sigma_z. \tag{29}$$

Having applied the Fourier transform (10) to Eqs. (26)–(29) and boundary and finiteness conditions (5) and (9), respectively, we can obtain the following boundary-value problem:

$$\bar{\Delta}_1\bar{\sigma}_{rz}(\rho)=-\frac{is}{1+\nu}\frac{d}{d\rho}\left(\bar{\sigma}(\rho)+\alpha(\rho)E\bar{T}(\rho)\right),\qquad \bar{\Delta}_2\left(\frac{1-\nu}{E}\bar{\sigma}(\rho)+2\alpha(\rho)\bar{T}(\rho)\right)=0, \tag{30}$$

$$\rho^2\bar{\sigma}_r(\rho)=\int\limits_0^\rho\xi\left(\bar{\sigma}(\xi)-\bar{\sigma}_z(\xi)-is\xi\bar{\sigma}_{rz}(\xi)\right)d\xi, \tag{31}$$

$$\bar{\sigma}_z(\rho)=\frac{i}{s\rho}\frac{d}{d\rho}\left(\rho\bar{\sigma}_{rz}(\rho)\right), \tag{32}$$

$$\bar{\sigma}_\varphi=\bar{\sigma}-\bar{\sigma}_r-\bar{\sigma}_z, \tag{33}$$

$$|\bar{\sigma}_r(0)|<\infty,\quad |\bar{\sigma}_{rz}(0)|<\infty,\quad \bar{\sigma}_r(1)=-\bar{p},\quad \bar{\sigma}_{rz}(1)=\bar{q}. \tag{34}$$

Putting Eqs. (32) into (31) yields

$$\rho^2\bar{\sigma}_r(\rho)+\frac{i\rho}{s}\bar{\sigma}_{rz}(\rho)=\int\limits_0^\rho\left(\xi\bar{\sigma}(\xi)-is\xi^2\bar{\sigma}_{rz}(\xi)\right)d\xi. \tag{35}$$

In view of boundary conditions (34), the latter equation takes the form of the following integral condition:

$$\int_0^1 \left(\xi\bar{\sigma}(\xi) - is\xi^2\bar{\sigma}_{rz}(\xi)\right) d\xi = \frac{i}{s}\bar{q} - \bar{p}. \tag{36}$$

In such manner, the original thermoelasticity boundary-value problem has been reduced to the determination of key stresses $\bar{\sigma}$ and $\bar{\sigma}_{rz}$ from the governing Eq. (30) accopmanied with the finiteness and boundary conditions (34) for the shearing stress and the integral condition (36). After the key stresses are determined, the remaining axial, radial, and circumferential stresses can be computed by means of Eqs. (32), (35), and (33), respectively.

The solution to the second of Eqs. (30) can be given, in view of finiteness conditions (34), in the form

$$\bar{\sigma}(\rho) = \frac{E}{1-\nu}\left(AI_0(s\rho) - 2\alpha(\rho)\bar{T}(\rho)\right), \tag{37}$$

where A is a constant of integration. By substituting the obtained expression (37) into the first of Eqs. (30) and solving it with respect to the shearing stress, we can obtain

$$\bar{\sigma}_{rz}(\rho) = BI_1(s\rho) - \frac{A}{2}\frac{iE}{1-\nu^2}\left(s\rho I_0(s\rho) - 2I_1(s\rho)\right) + \frac{is^2E}{1-\nu}t_{10}(\rho). \tag{38}$$

Here, B is a constant of integration and function $t_{10}(\rho)$ is expressed as

$$t_{lm}(\rho) = \int_0^\rho \xi\alpha(\xi)\bar{T}(\xi)\chi_{lm}(\rho,\xi)d\xi, \tag{39}$$

where $\chi_{lm}(\rho,\xi)$ is given by formula (14).

By substituting expressions (37) and (38) into the last boundary condition (34) and the integral condition (36), we can obtain two algebraic equations for the determination of constants A and B, so that the shearing and total stresses are thereby determined. Then, by taking formulae (32), (35), and (33) into account, we arrive at the following expression

$$\bar{\sigma}_\tau(\rho) = \bar{p}f_p^\tau(\rho) + \bar{q}f_q^\tau(\rho) + f_T^\tau(\rho) \tag{40}$$

for the stress-tensor components, where $\tau = \{r, \varphi, z, rz\}$ and the functions $f_p^\tau(\rho)$, $f_q^\tau(\rho)$, and $f_T^\tau(\rho)$ are given in the Appendix.

Putting the obtained expressions (40) for the stress-tensor components into the constitutive equations (6), we can determine the strains. Then, by making use of the Cauchy relations (7), the displacements can be found in the form

$$u_\tau = \bar{p}g_p^\tau(\rho) + \bar{q}g_q^\tau(\rho) + g_T^\tau(\rho), \tag{41}$$

where $\tau = \{r, z\}$, and functions $g_p^\tau(\rho)$, $g_q^\tau(\rho)$, and $g_T^\tau(\rho)$ are given in the Appendix.

3.2.2 The case of given displacements on the boundary

In the case of the boundary conditions in terms of displacements (8), the problem on determination of thermal stresses and displacements in the cylinder can be reduced to the solution considered in previous section. Using $\rho = 1$ in (41) with (8) in mind yields the system of two equations

$$\begin{cases} \bar{p}g_p^r(1) + \bar{q}g_q^r(1) = \bar{u} - g_T^r(1), \\ \bar{p}g_p^z(1) + \bar{q}g_q^z(1) = \bar{w} - g_T^z(1). \end{cases} \tag{42}$$

Due to the fact that the determinant of this system is not equal to zero, the boundary tractions $\bar{p}$ and $\bar{q}$ can be determined from this system in the form

$$\bar{p} = \frac{\left(\bar{u} - g_T^r(1)\right)g_q^z(1) - \left(\bar{w} - g_T^z(1)\right)g_q^r(1)}{g_p^r(1)g_q^z(1) - g_p^z(1)g_q^r(1)}, \qquad \bar{q} = \frac{\left(\bar{w} - g_T^z(1)\right)g_p^r(1) - \left(\bar{u} - g_T^r(1)\right)g_p^z(1)}{g_p^r(1)g_q^z(1) - g_p^z(1)g_q^r(1)}. \tag{43}$$

In this manner, by substituting the expressions (43) into formulae (40) and (41), the stresses and displacements can be determined for the boundary conditions (8).

Note that system (42) can be used for the solution of thermoelasticity problem for mixed-type boundary conditions, when one component of the stress tensor and one component of the displacement vector are given on the boundary $\rho = 1$.

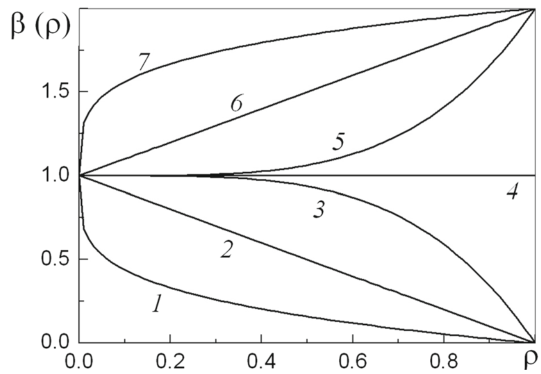

Fig. 1 Function β_x versus ρ for $m_x = -1, l_x = 1/4$ (curve 1), $m_x = -1, l_x = 1$ (curve 2), $m_x = -1, l_x = 4$ (curve 3), $m_x = 0$, (curve 4), $m_x = 1, l_x = 4$ (curve 5), $m_x = 1, l_x = 1$ (curve 6), $m_x = 1, l_x = 1/4$ (curve 7)

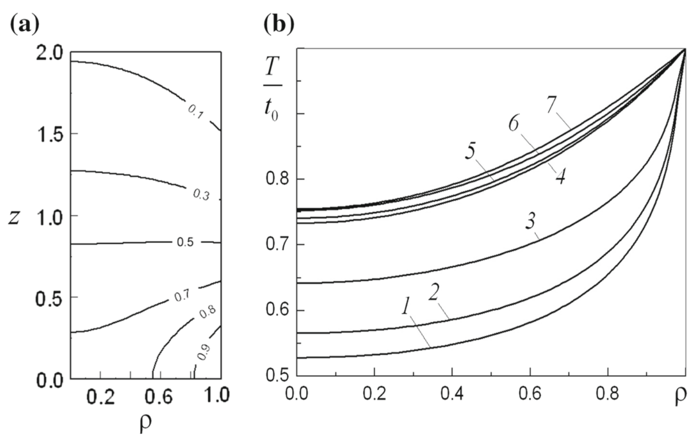

Fig. 2 **a**—Full-field distribution of the dimensionless temperature $T(\rho, z)/t_0$ for $m_\lambda = 0$ (homogeneous material) due to thermal loading (45); **b**—ρ-distribution of the dimensionless temperature at $z = 0$ for: $m_\lambda = -1, l_\lambda = 1/4$ (curve 1), $m_\lambda = -1, l_\lambda = 1$ (curve 2), $m_\lambda = -1, l_\lambda = 4$ (curve 3), $m_\lambda = 0$, (curve 4), $m_\lambda = 1, l_\lambda = 1/4$ (curve 5), $m_\lambda = 1, l_\lambda = 1$ (curve 6), $m_\lambda = 1, l_\lambda = 4$ (curve 7)

4 Numerical examples and discussion

Consider the computation of temperature, thermal stresses and displacements in a cylinder with the thermo-physical properties

$$\lambda(\rho) = \lambda_0 \beta_\lambda(\rho), \qquad \alpha(\rho) = \alpha_0 \beta_\alpha(\rho), \qquad \beta_x(\rho) = 1 + m_x \rho^{l_x}, \quad x = \{\lambda, \alpha\}, \tag{44}$$

where λ_0 and α_0 are the values of moduli on the axis of the cylinder, m_x and l_x are real numbers, $x = \{\lambda, \alpha\}$. The distribution of function $\beta_x(\rho)$ is shown in Fig. 1 for $m_x = 0; \pm 1$ and $l_x = 0; 1; 4; 1/4$. For $m_x < 0$, the moduli are decreasing functions of ρ and, for the considered value $m_x = -1$, equal zero on the lateral surface (in this case, the surface is assumed to be non-conductive and non-thermal-expandable). For $m_x > 0$, the moduli have their minimum on the axis of the cylinder and attain their maximum on the lateral surface. The case $m_x = 0$ corresponds to the constant moduli (homogeneous material). If $l_x < 1$, then the moduli exhibit greater variation in the vicinity of the axis. If $l_x > 1$, then the variation occurs near to the lateral surface.

Let the cylinder be subjected to the external heating (2), where

$$T_0 = t_0 \exp(-t_1 z^2), \quad a = 1, \quad b = 0, \tag{45}$$

t_0 is a constant parameter in the dimension of temperature and t_1 is a positive real number.

In Fig. 2, the distribution of dimensionless temperature $T(\rho, z)/t_0$ due the thermal loading (45) (here and in what follows $t_1 = 1$) is shown for the heat-conduction coefficient (44) corresponding to the case of homogeneous material ($m_\lambda = 0$) and inhomogeneous ones for different values of m_λ and l_λ. For the case

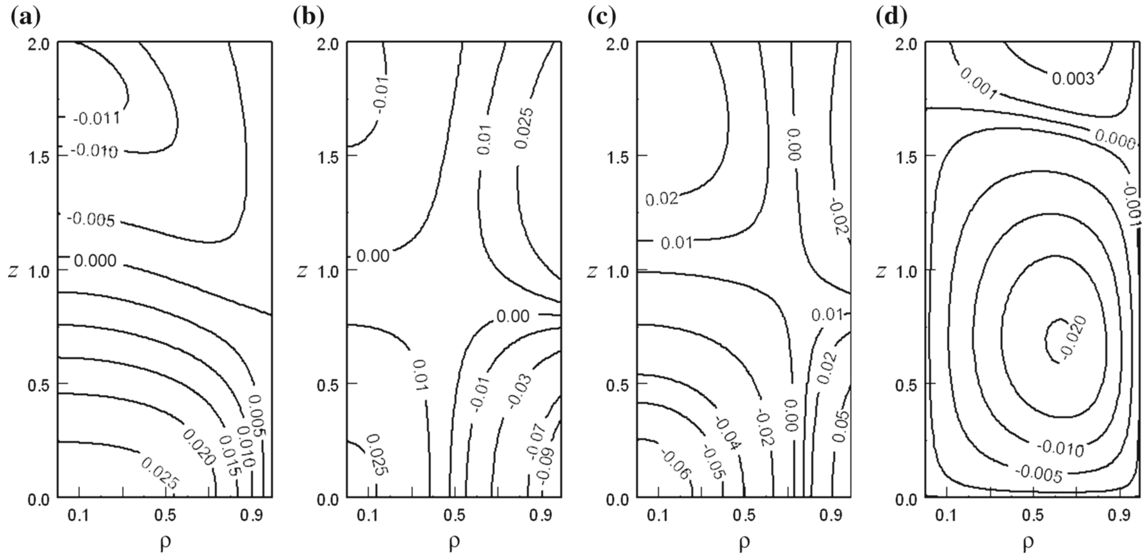

Fig. 3 Full-field distributions of the dimensionless (normalized by $\alpha_0 E t_0$) radial **a**, circumferential **b**, axial **c**, and shearing **d** stress-tensor components due to thermal loading (45) of the cylinder with the stress-free boundary ($p = q = 0$) for in the case of homogeneous material ($m_\lambda = m_\alpha = 0$)

of inhomogeneous material, the computations have been provided on the basis of formula (23) for $N = 4$, which provides sufficient engineering accuracy, and, alternatively, the quadrature scheme (17) with $M = 100$, which showed perfect agreement with the afore-mentioned analytical result. Due to the loading (45), the distribution of temperature is symmetric with respect to the cross section $z = 0$. For both homogeneous and inhomogeneous materials, the maximum temperature naturally occurs on the lateral surface at the zone of maximum thermal loading (45) and decreases when approaching the cylinder axis along the radius. In the case of inhomogeneous material (Fig. 2b), the distributions of temperature for $m_\lambda > 0$ nearly coincide with the distribution in homogeneous material, which allows us to conclude that the character of increment in the heat-conduction coefficient in the form (44) with the increment of radial coordinate does not play the significant role in the distribution of temperature. In contrast, the heat-conduction coefficient for $m_\lambda < 0$ effects in the rapid decrement of the temperature when moving away from the lateral surface (Fig. 2b, curves 1, 2, 3). The smaller the parameter l_λ, the lower is the temperature in the vicinity of the cylinder axis.

The full-filed distributions of the dimensionless thermal stresses computed by means of formula (40) in the case of homogeneous material ($m_\lambda = m_\alpha = 0$) are shown in Fig. 3 for the stress-free lateral surface ($p = q = 0$). Due to the distribution of temperature computed by formula (20), the normal stresses are symmetric functions of the coordinate z, while the shearing stress is an antisymmetric function of this coordinate. All stresses vanish when moving away from the heated part of the cylinder in the axial direction (for $|z| >> 1$). Due to the fact that the radial stress (Fig. 3a) equals zero on the boundary, it attains its maximum tensile value on the cylinder axis at $z = 0$. When moving away from this point along the axis, the stress decreases and becomes compressive for $|z| > 1$. The circumferential stress (Fig. 3b) peaks its maximum compressive value on the lateral surface at the points of maximum temperature and, when moving away from this point in both radial and axial direction, this stress becomes tensile. In the zone $z > 1.5$ and $r < 0.3$, this stress becomes compressive. The axial stress (Fig. 3c) exhibits the exact opposite qualitative behavior. The shearing stress (Fig. 3d) attains the maximum absolute values in the vicinity of point $(0.65, 0.55)$ and equals zero on the lateral surface (owing to the boundary condition), as well as at $\rho = 0$ and $z = 0$ (owing to the symmetry).

The inhomogeneity effect is demonstrated in Figs. 4 and 5 depicting the radial and axial stresses, respectively, in the characteristic sections of the cylinder $z = 0$, $\rho = 0$ and $\rho = 1$. As we can observe, the variation of the thermal expansion coefficient within the radial coordinate effects significantly in the stress distribution. In the case, when the coefficient suffers the greater variation near to the lateral surface ($m_\alpha < 1$), the maximum of the radial and axial stresses on the cylinder axis is smaller in comparison to the case, when the linear thermal coefficient varies rapidly near to the axis ($m_\alpha > 1$). Note that in the latter case, the coefficient $\alpha(\rho)$ is non-smooth on the axis, which effects in significant rise of the stresses. The radial distribution of the axial stress (Fig. 5a) allows us to conclude that this stress meets the condition

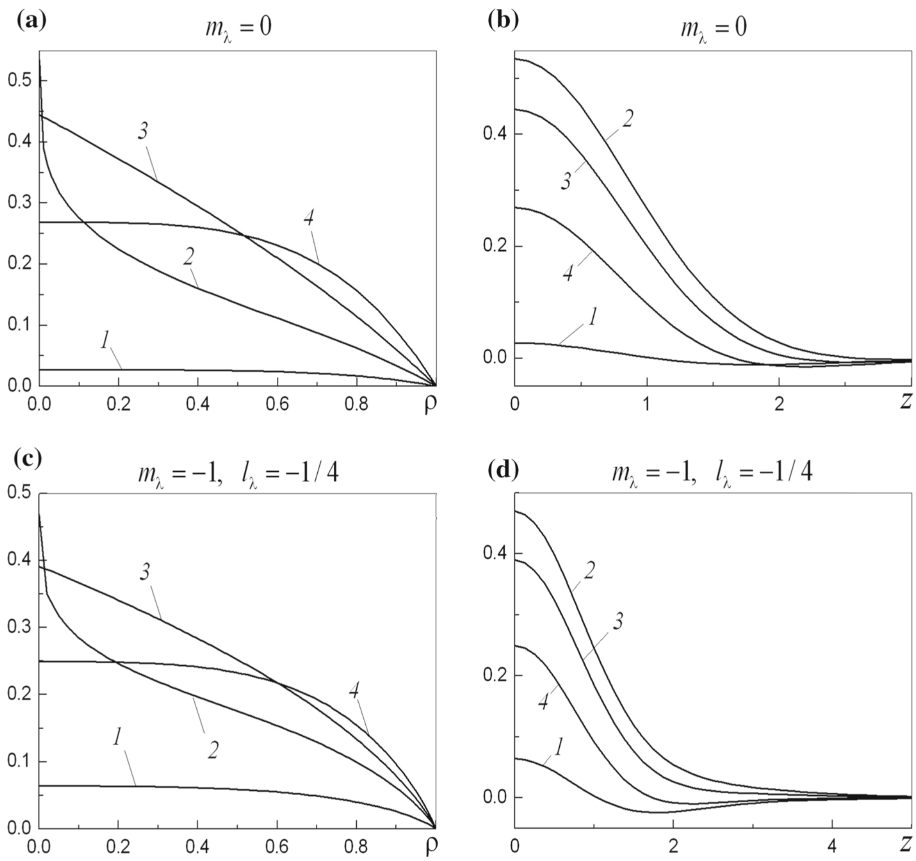

Fig. 4 ρ−Distribution at $z = 0$ **a, c** and z−distribution at $\rho = 0$ **b, d** of the radial stress $\sigma_r/(\alpha_0 E t_0)$ for $p = q = 0$, $m_\lambda = 0$ **a, b** and $m_\lambda = -1$, $l_\lambda = 1/4$ **c, d**, and different values of the inhomogeneity parameters: $m_\alpha = 0$, (curve 1), $m_\alpha = 1$, $l_\alpha = 1/4$ (curve 2), $m_\alpha = 1$, $l_\alpha = 1$ (curve 3), $m_\alpha = 1$, $l_\alpha = 4$ (curve 4)

$$\int_0^1 \rho \sigma_z(\rho, z) d\rho = 0,$$

which follows from the equilibrium equations (4) and homogeneous boundary condition for the shearing stress.

The dimensionless displacements $u_r(\rho, z)/(\alpha_0 t_0)$ and $u_z(\rho, z)/(\alpha_0 t_0)$ due to thermal loading (45) are depicted in Fig. 6 for the case of homogeneous material ($m_\lambda = m_\alpha = 0$). As we can observe, the radial displacement peaks its maximum value in the center of most heated part of the lateral surface and is equivalent to zero on the cylinder axis. The axial displacement is an antisymmetric function of the axial coordinate. It tends to a constant value for $z > 1$, which corresponds to the finite extension of the material in axial direction due to heating of the force-free boundary.

Figure 7 presents the radial displacement versus ρ at $z = 0$ (Fig. 7a) and versus z on the lateral surface (Fig. 7b) for different values of the parameter l_α. Due to the fact that the radial displacement equals zero on the cylinder axis, the significant effect of inhomogeneity is observed on the lateral surface. For all cases of inhomogeneity, the displacement profile on the lateral surface (Fig. 7b) resembles qualitatively the profile of thermal loading (45).

The axial displacement on the cylinder axis and lateral surface is shown in Fig. 8 versus coordinate z. As we can conclude from this figure, the radial dependence of the linear thermal expansion coefficient effects in the mismatch of the constant value, to which the axial displacement tends when moving away from the thermally affected part along the cylinder axis, for different cases of inhomogeneity.

Consider the case, when the boundary displacements u and w are given on the boundary (8) instead of the boundary tractions p and q in (5). Let the lateral surface of the cylinder with material properties (44), which is subjected to thermal loading (45), be rigidly fixed, i.e.

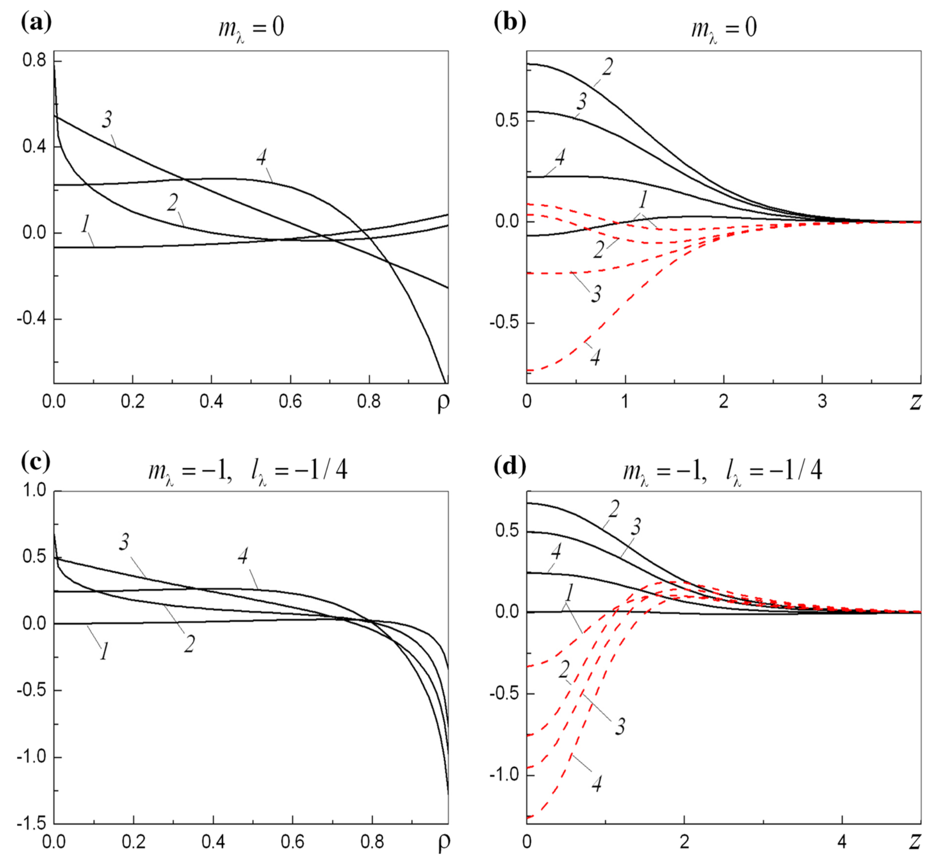

Fig. 5 $\rho-$Distribution at $z = 0$ **(a, c)** and $z-$distribution at $\rho = 0$ (solid lines) and $\rho = 1$ (dashed lines) **(b, d)** of the axial stress $\sigma_z/(\alpha_0 E t_0)$ for $p = q = 0$, $m_\lambda = 0$ **(a, b)** and $m_\lambda = -1$, $l_\lambda = -1/4$ **(c, d)**, and different values of the inhomogeneity parameters: $m_\alpha = 0$, (curve 1), $m_\alpha = 1$, $l_\alpha = 1/4$ (curve 2), $m_\alpha = 1$, $l_\alpha = 1$ (curve 3), $m_\alpha = 1$, $l_\alpha = 4$ (curve 4)

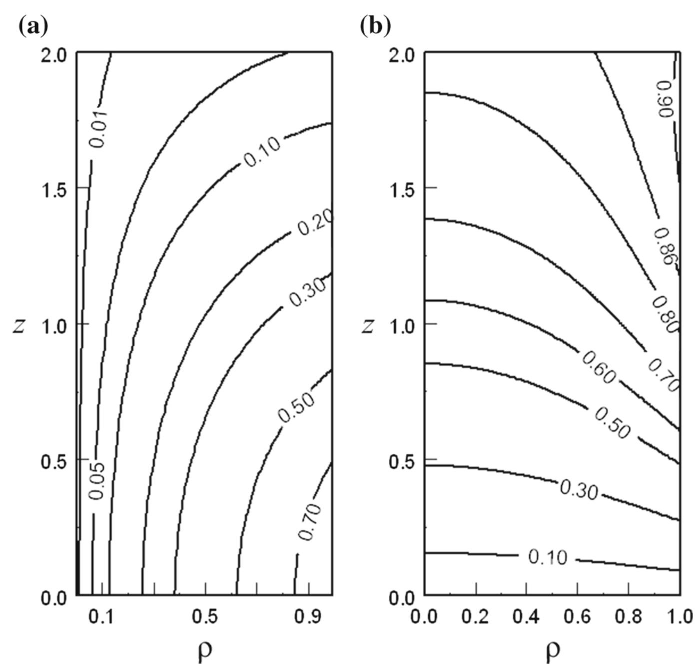

Fig. 6 Full-field distributions of the dimensionless radial **a** and axial **b** displacements in the case of homogeneous material ($m_\lambda = m_\alpha = 0$) due to thermal loading (45) for $p = q = 0$

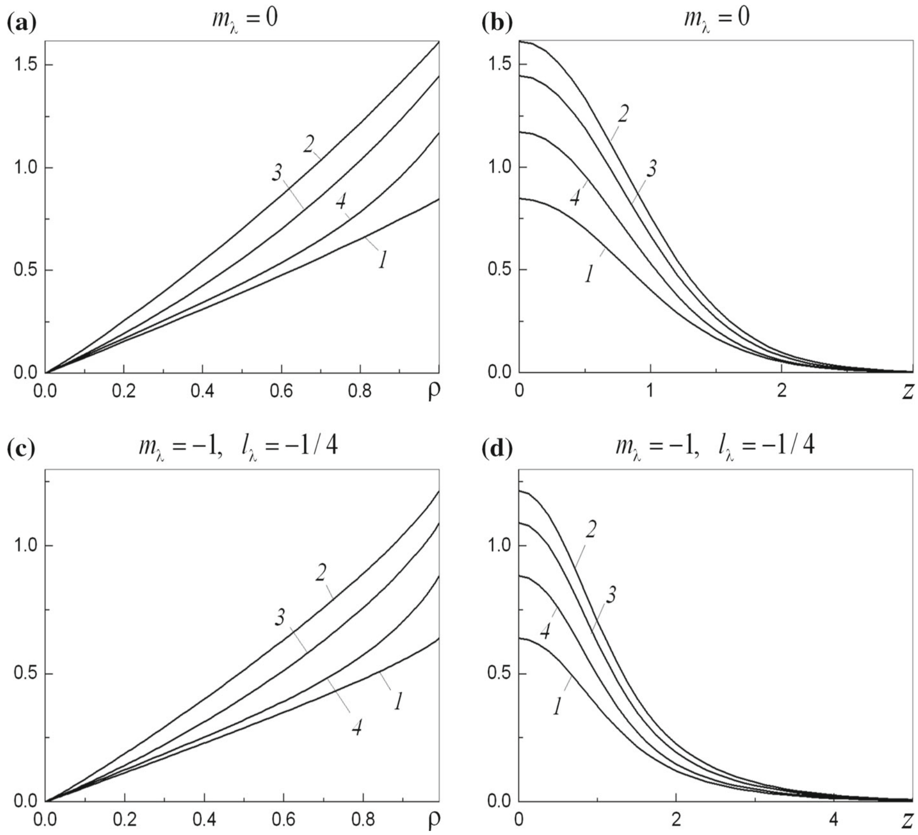

Fig. 7 $\rho-$Distribution at $z = 0$ **a, c** and $z-$distribution at $\rho = 1$ **b, d** of the radial displacement $u_r/(\alpha_0 t_0)$ for $p = q = 0$, $m_\lambda = 0$ **a, b** and $m_\lambda = -1$, $l_\lambda = 1/4$ **c, d** and different values of the inhomogeneity parameters: $m_\alpha = 0$, (curve 1), $m_\alpha = 1$, $l_\alpha = 1/4$ (curve 2), $m_\alpha = 1$, $l_\alpha = 1$ (curve 3), $m_\alpha = 1$, $l_\alpha = 4$ (curve 4)

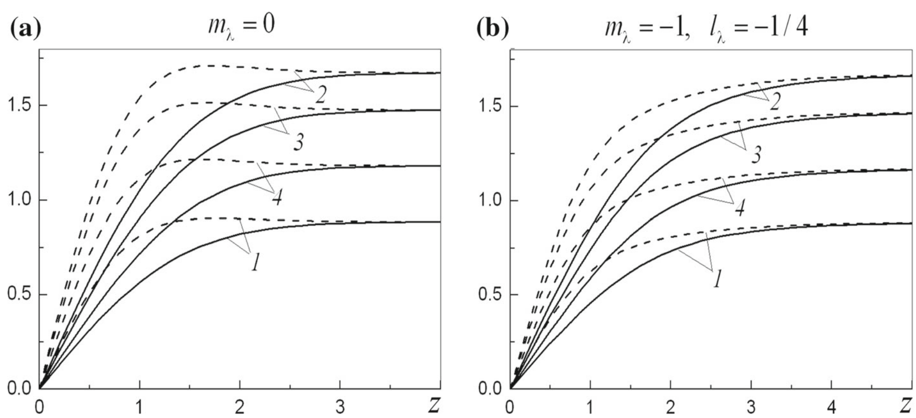

Fig. 8 The axial displacement $u_z/(\alpha_0 t_0)$ on the axis of the cylinder (solid lines) and lateral surface (dashed lines) for $p = q = 0$, $m_\lambda = 0$ **a** and $m_\lambda = -1$, $l_\lambda = 1/4$ **b** at $m_\alpha = 0$, (curve 1), $m_\alpha = 1$, $l_\alpha = 1/4$ (curve 2), $m_\alpha = 1$, $l_\alpha = 1$ (curve 3), $m_\alpha = 1$, $l_\alpha = 4$ (curve 4)

$$u = w = 0, \tag{46}$$

Then, by making use of the expressions (43), we can determine the boundary tractions p and q and compute the stresses (40) and displacements (41) in the case of boundary conditions (8) and (46).

The full-field distributions of the stress-tensor and displacement-vector components which correspond to the boundary conditions (46) are depicted in Fig. 9 for the case of homogeneous material. In contrast to the case of force-free boundary (Fig. 3), all stress-tensor components have nonzero distributions on the lateral

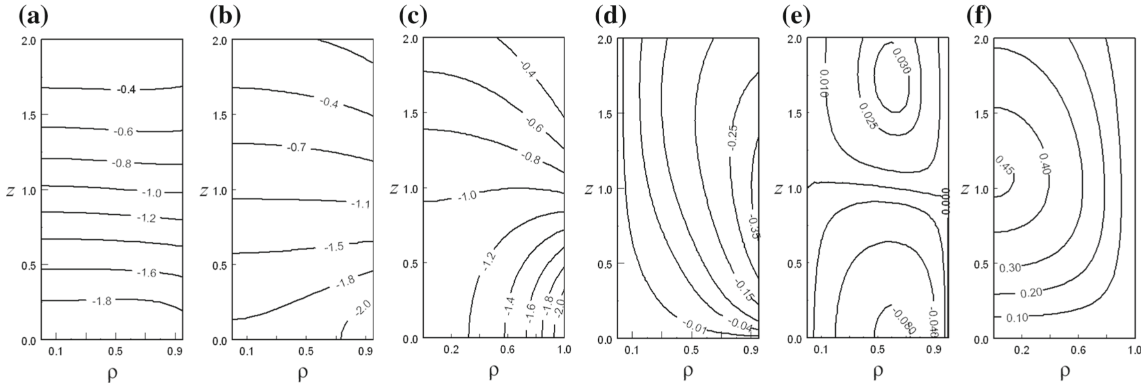

Fig. 9 Full-field distributions of the dimensionless radial **a**, circumferential **b**, axial **c**, and shearing **d** stress-tensor components and radial **e** and axial **f** displacements due to thermal loading (45) of the cylinder with the fixed boundary ($u = w = 0$) for in the case of homogeneous material ($m_\lambda = m_\alpha = 0$)

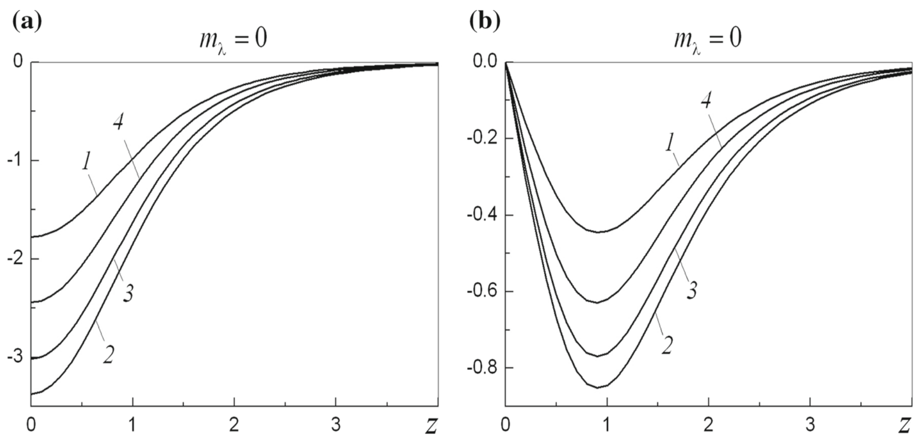

Fig. 10 The radial (***a***) and shearing (***b***) stresses on the fixed lateral surface $\rho = 0$ for $m_\lambda = 0$ at $m_\alpha = 0$, (curve *1*), $m_\alpha = 1$, $l_\alpha = 1/4$ (curve *2*), $m_\alpha = 1$, $l_\alpha = 1$ (curve *3*), $m_\alpha = 1$, $l_\alpha = 4$ (curve *4*)

surface. The radial (Fig. 9a) and circumferential (Fig. 9b) stresses are nearly uniform with respect to the radial coordinate and vanish when moving away from the thermally effected zone. The axial stress (Fig. 9c) attains its maximum on the lateral surface at the point of the maximum applied temperature. The shearing stress (Fig. 9d) peaks its maximum on the lateral surface on a distance from the maximum of temperature, which approximately equals the cylinder radius. At the central cross section $z = 0$, the radial displacement (Fig. 9e) attains its maximum for $\rho \approx 0.6$ and is negative throughout the half-thickness of the cylinder. On the distance from $z = 0$, which equals the radius, this displacement switch from negative to positive, however, of lesser magnitude. When moving away from that zone, the displacement vanishes. The axial displacement (Fig. 9f) attains its maximum on the cylinder axis on a distance from the central cross section, which approximately equals the radius. In contrast to the case of force-free boundary (see Figs. 6b and 8), this displacement vanish for greater values of z.

The effect of inhomogeneity in the distribution of the radial and shearing tractions on the fixed lateral surfaces is depicted in Fig. 10. The qualitative behavior of the radial traction resembles the applied thermal loading. The shearing traction attains its maximum (of different magnitude) in the location with the same coordinate z for different values of the parameters of inhomogeneity.

Note that the distributions of stresses and displacements for $m_\alpha < 0$ are symmetric to the ones for $m_\alpha > 0$ with respect to the distributions in the homogeneous material, due to the dependence (44) and the fact that the stresses depend linearly on the coefficient $\alpha(\rho)$.

5 Conclusions

By making use of the direct integration method, we constructed the exact analytical solutions to the axisymmetric heat-conduction and thermoelasticity problems for a long solid cylinder with arbitrary variation of its thermophysical properties (the linear thermal expansion coefficient and the heat-conduction coefficient) with the radial coordinate. The stationary heat-conduction problem has been reduced to an integral equation of the second kind. For the solution of the latter equation, the numerical, analytical-numerical and analytical methods have been proposed. By making use of the resolvent-kernel technique, the temperature has been determined in an explicit analytical form. Having determined the temperature, the original thermoelasticity problem with boundary conditions in terms of stresses has been reduced to a set of two governing equations for two key functions, which are the total and shearing stresses. The boundary condition for the radial stress was equivalently reduced to an integral-boundary condition for the key functions, which allowed for the completing of the boundary-value problem for the mentioned functions. Having determined the key functions from the latter problem, we found the stress-tensor components by making use of their expressions through the key functions, which have been established by the integration of the equilibrium equations.

For the case when the displacements or mixed conditions are imposed on the boundary, the one-to-one relations were established for the determination of the boundary tractions in terms of the given displacements. Along with the fact that the solution for the temperature has been obtained in an explicit form, this allowed for the reduction of the problem to the one in terms of stresses, whose solution has already been obtained.

The numerical analysis showed that in the case of force-free lateral surface, the distribution of radial displacement on the heated part of the boundary resembles qualitatively the profile of applied thermal loading. If the lateral surface is rigidly fixed, the same can be concluded for the radial traction.

Appendix

$$
\begin{aligned}
f_p^r(\rho) &= \frac{1}{\rho\mu(1)}\Big(s^2\left(1-\rho^2\right) I_1(s) I_1(s\rho) - s\vartheta_1(\rho) - \mu(\rho)\Big),\\
f_q^r(\rho) &= \frac{i}{\rho\mu(1)}\Big((1-2\nu)\vartheta_1(\rho) + s\left(1-\rho^2\right) I_1(s) I_1(s\rho) - s^2\rho\vartheta_2(\rho)\Big),\\
f_T^r(\rho) &= \frac{s}{\rho\mu(1)}\frac{E}{1-\nu}\Big(\mu(1)\left(s\rho t_{00}(\rho) - t_{10}(\rho)\right) + \left(s^3\left(1-\rho^2\right) I_1(s) I_1(s\rho) - s^2\vartheta_1(\rho) - s\mu(\rho)\right) t_{00}(1)\\
&\quad + \left(\mu(\rho) + 2s(1-\nu)\vartheta_1(\rho) - s^3\rho\vartheta_2(\rho)\right) t_{10}(1)\Big), \qquad f_p^\varphi(\rho) = \frac{1}{\rho\mu(1)}\Big(s\vartheta_1(\rho) + 2I_1(s)\vartheta(\rho)\Big),\\
f_q^\varphi(\rho) &= \frac{i}{\rho\mu(1)}\Big(\vartheta_1(\rho) - 2I_0(s)\vartheta(\rho) - s\vartheta_4(\rho) + 2\nu\rho I_1(s) I_0(s\rho)\Big),\\
f_T^\varphi(\rho) &= \frac{s}{\rho\mu(1)}\frac{E}{1-\nu}\Big(\mu(1)\left(t_{10}(\rho) - \rho s^{-1}\alpha(\rho)\bar{T}(\rho)\right) - \left(\mu(\rho) + 2sI_0(s)\vartheta(\rho)\right) t_{10}(1)\\
&\quad + s\left(s\vartheta_1(\rho) + 2I_1(s)\vartheta(\rho)\right) t_{00}(1)\Big), \qquad f_p^z(\rho) = \frac{s}{\mu(1)}\Big(2I_1(s)I_0(s\rho) - s\vartheta_3(\rho)\Big),\\
f_q^z(\rho) &= \frac{i}{\mu(1)}\Big(s^2\vartheta_2(\rho) + 2\left((2-\nu)I_1(s) - sI_0(s)\right) I_0(s\rho) - s\vartheta_3(\rho)\Big),\\
f_T^z(\rho) &= \frac{s}{\mu(1)}\frac{E}{1-\nu}\Big(s\left(s^2\vartheta_2(\rho) + 2\left((1-\nu)I_1(s) - sI_0(s)\right) I_0(s\rho)\right) t_{10}(1)\\
&\quad + s^2\left(2I_1(s)I_0(s\rho) - s\vartheta_3(\rho)\right) t_{00}(1) - \mu(1)\left(s^{-1}\alpha(\rho)\bar{T}(\rho) + st_{00}(\rho)\right)\Big),\\
f_p^{rz}(\rho) &= \frac{is^2}{\mu(1)}\vartheta_1(\rho), \qquad f_q^{rz}(\rho) = \frac{1}{\mu(1)}\left(\mu(\rho) - s\vartheta_1(\rho)\right),\\
f_T^{rz}(\rho) &= \frac{is^2}{\mu(1)}\frac{E}{1-\nu}\left(\mu(1)t_{10}(\rho) - \mu(\rho)t_{10}(1) + s^2\vartheta_1(\rho)t_{00}(1)\right),\\
g_p^r(\rho) &= \frac{1+\nu}{\mu(1)E}\Big(s\vartheta_1(\rho) + 2(1-\nu)I_1(s)I_1(s\rho)\Big),
\end{aligned}
$$

$$
\begin{aligned}
g_q^r(\rho) &= i\frac{1+\nu}{\mu(1)E}\Big(\vartheta_1(\rho) - s\vartheta_4(\rho) - 2(1-\nu)I_0(s)I_1(s\rho)\Big),\\
g_T^r(\rho) &= \frac{s}{\mu(1)}\frac{1+\nu}{1-\nu}\Big(\mu(1)t_{10}(\rho) + s\,(s\vartheta_1(\rho) + 2(1-\nu)I_1(s)I_1(s\rho))\,t_{00}(1)\\
&\quad - (\mu(\rho) + 2s(1-\nu)I_0(s)I_1(s\rho))\,t_{10}(1)\Big),\\
g_p^z(\rho) &= i\frac{1+\nu}{\mu(1)E}\Big(s\vartheta_3(\rho) - 2(1-\nu)I_1(s)I_0(s\rho)\Big),\\
g_q^z(\rho) &= \frac{1+\nu}{s\mu(1)E}\Big(s^2\vartheta_2(\rho) - s\vartheta_3(\rho) + 2(1-\nu)\,(2I_1(s) - sI_0(s))\,I_0(s\rho)\Big),\\
g_T^z(\rho) &= \frac{is}{\mu(1)}\frac{1+\nu}{1-\nu}\Big(\mu(1)t_{00}(\rho) + s\,(s\vartheta_3(\rho) - 2(1-\nu)I_1(s)I_0(s\rho))\,t_{00}(1)\\
&\quad - \big(s^2\vartheta_2(\rho) + 2(1-\nu)(I_1(s) - sI_0(s))I_0(s\rho)\big)\,t_{10}(1)\Big).
\end{aligned}
$$

Here, $t_{00}(\rho)$ and $t_{10}(\rho)$ are given by formula (39) and

$$
\begin{aligned}
&\vartheta_1(\rho) = I_0(s)I_1(s\rho) - \rho I_1(s)I_0(s\rho), \quad \vartheta_2(\rho) = I_1(s)I_0(s\rho) - \rho I_0(s)I_1(s\rho),\\
&\vartheta_3(\rho) = I_0(s)I_0(s\rho) - \rho I_1(s)I_1(s\rho), \quad \vartheta_4(\rho) = I_1(s)I_1(s\rho) - \rho I_0(s)I_0(s\rho),\\
&\vartheta(\rho) = \nu s\rho I_0(s\rho) + (1-\nu)I_1(s\rho), \quad \mu(\rho) = \big(s^2 + 2 - 2\nu\big)\,I_1(s)I_1(s\rho) - s^2\rho I_0(s)I_0(s\rho).
\end{aligned}
$$

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